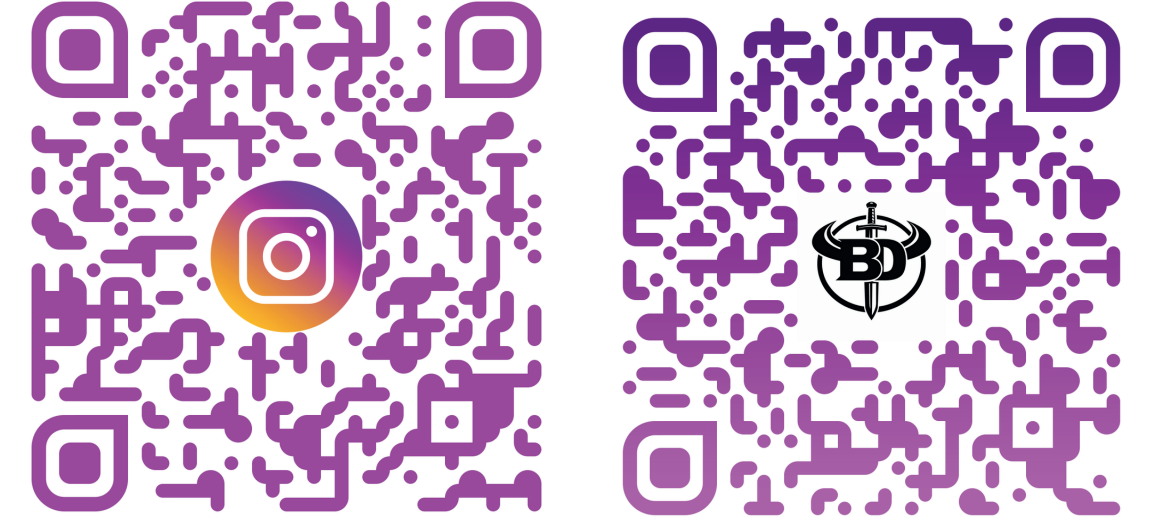


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BODENSEE DEVILS

Roboter AG Markdorf



Team

Tobias Schauer - Software
Tim Eickhoff - Hardware
Nils Marte - Hardware

Karl-Gustav der III - Robot

Challenges and Development

The biggest challenges for us were optimizing the ball approach and developing the dribbler. The ball approach is a challenge in both the software and the hardware. The best approach here was to use a simple state machine to apply various factors to the angle. On the hardware side, we adapted the ball holder so that it fits perfectly onto the ball and integrates a dribbler. This dribbler runs on a sensor, a brushless motor, and a silicone roller. Getting it to run with the driver was a very time-consuming challenge.

Software

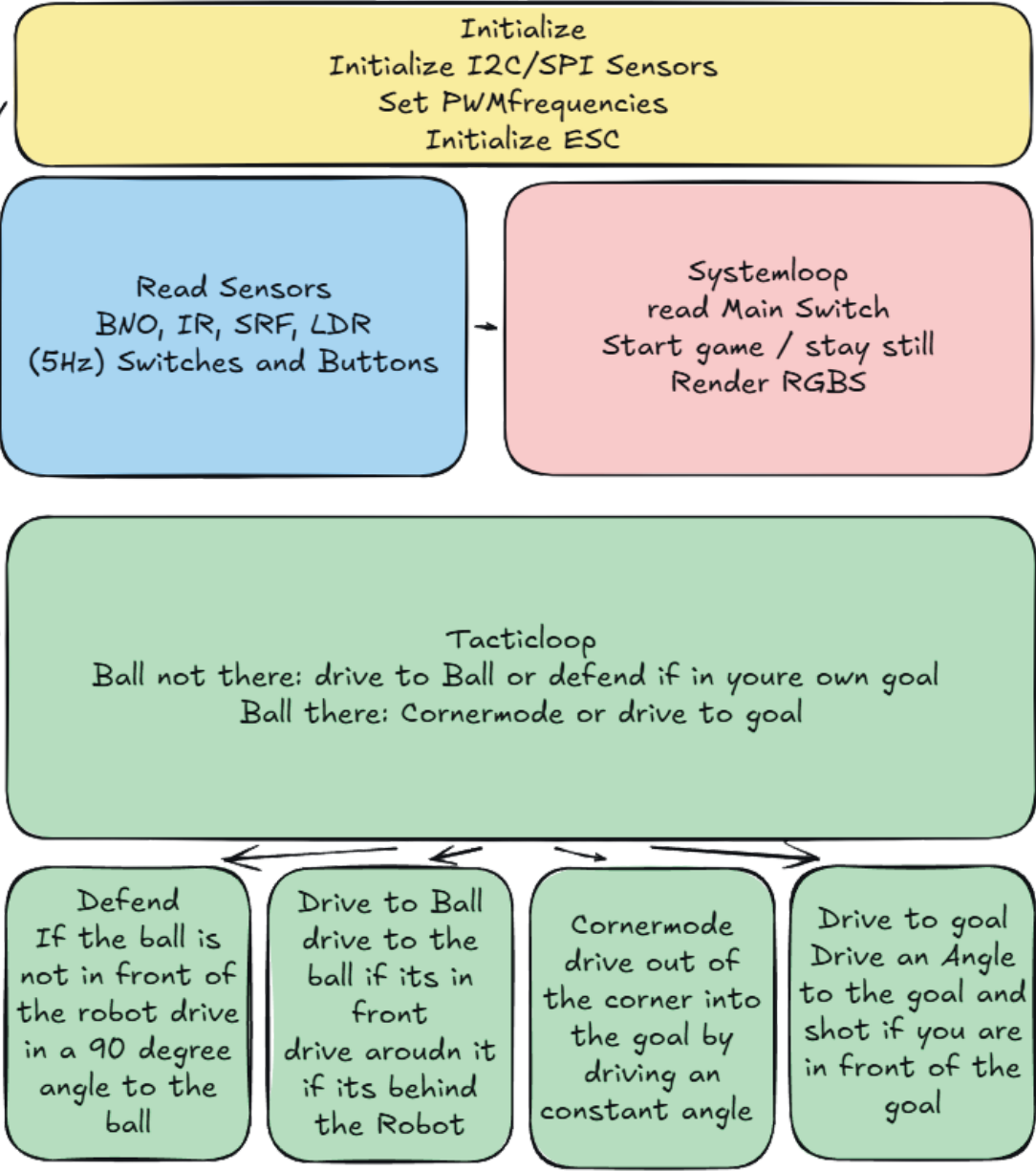
Robot Software on Teensy 4.0

Our robot's software runs on a Teensy 4.0 and has a cycle time of about 5 to 6 milliseconds. During this time, sensor data is read, the next steps are planned, and commands are sent to the motors. The program was developed in C/C++ using PlatformIO in Visual Studio Code.

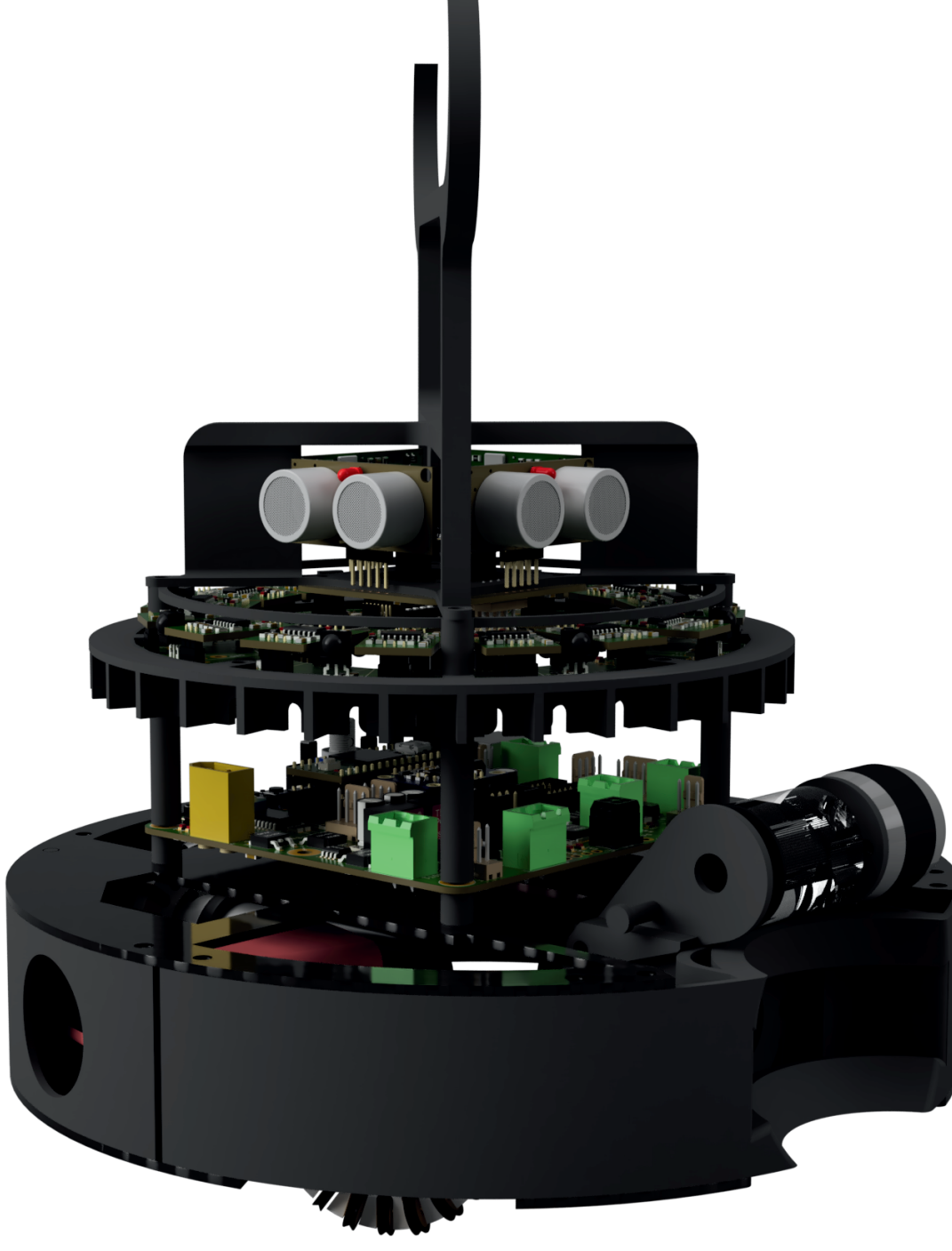
Main Logic and Tactics

The main logic consists of a comprehensive state machine that calls the appropriate tactics. When the robot does not have possession of the ball, we distinguish between defense and ball approach based on the ultrasonic values and the ball's angle. As soon as the robot has the ball, it either steers toward the center of the field to shoot or directly toward the goal. If the robot detects another robot or the corner, it moves diagonally backward toward the center of the field to pass the opponent.

complete source code:



Karl-Gustav der III



- Teensy 4.0:**
The heart of the robot is a Teensy 4.0, which collects and processes all sensor data at approximately 200 Hz and controls the motor drivers.
- IR board with STM32:**
Using 32 IR diodes and 16 operational amplifiers, which filter the ball's 40 kHz signal, an STM32 calculates the ball's angle and an approximate distance to the ball.
- BNO 055:**
We use a BNO 055 as a compass to control and correct our orientation via a PID controller.
- SRF08:**
The 4 ultrasonic sensors are used for position and opponent detection.
- LDR:**
A classic light barrier quickly and reliably informs the robot whether it is in possession of the ball.
- Chassis and Materials:**
The robot's chassis was designed in Fusion360 and largely 3D-printed. The plates were milled from carbon fiber for better weight distribution and greater stability.
- Movement:**
The robot is powered by three Faulhaber 12V motors with omni-wheels. The motors provided by our sponsor offer a particular advantage due to their power.
- Dribbler:**
We use a sensorless brushless drone motor as the dribbler. This gives us tremendous control over the ball.
- Kicker:** We power a 3V shot for a few milliseconds using 12V. This allows us to precisely adjust how hard the robot shoots.